

# H<sup>+</sup>-H<sub>2</sub>O collision: measured nuclear motion

1 keV incoming proton |  $b = 2$  bohr | Octopus Ehrenfest dynamics

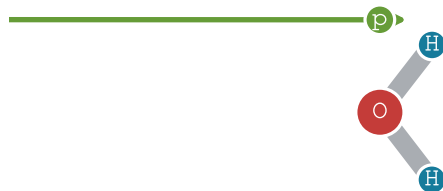
Approach

$t = 44.98$  a.u.



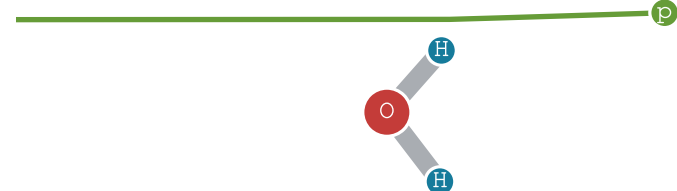
Closest approach to oxygen

$t = 74.95$  a.u.



Departure

$t = 105.03$  a.u.



Red: oxygen    Blue: water hydrogens    Green: projectile nucleus and past trajectory

Equal-scale laboratory xy projection. Atom sizes and bond strokes are illustrative.  
Nuclear positions come from the simulation; electron density and projectile charge state are not depicted.

One collision trajectory, three time samples. Numerical uncertainty has not yet been quantified.